

Climate change

For the last half century, Earth's climate has been getting steadily warmer. The world's climate has always varied naturally, but the evidence suggests that this warming is caused by human activity—and it could have a huge impact on our lives.

Humans make the world warmer mainly by burning fossil fuels such as coal and oil, which fill the air with carbon dioxide that traps the sun's heat. This is often referred to as global warming, but scientists prefer to talk about climate change because the unpredictable effects include fueling extreme weather. In future, we can expect more powerful storms and flooding as well as hotter summers and droughts.

Greenhouse effect

The cause of global warming is the greenhouse effect. In the atmosphere, certain gases—known as greenhouse gases—absorb heat radiation that would otherwise escape into space. This causes our planet to be warmer than it would be if it had no atmosphere. The main greenhouse gases are carbon dioxide, methane, nitrous oxide, and water vapor.

2 Reflection

Almost a third of the energy in sunlight is reflected back into space as UV and visible light.

Transportation

Gasoline- and diesel-guzzling trucks and cars, as well as fuel-burning airplanes, produce around 15 percent of greenhouse gases.

Industry

Heavy industry burning fossil fuels for energy adds about 13 percent of global greenhouse gas emissions.

Farming and deforestation

Intensively farmed cows, sheep, and goats release huge amounts of methane, a greenhouse gas. Forests absorb carbon dioxide, so deforestation leaves more carbon dioxide in the atmosphere.

1 Light from the sun

The sunlight that passes through the atmosphere is a mixture of types of radiation: ultraviolet (UV—short wave), visible light (medium wave), and infrared (long wave).

Power stations

Burning coal, natural gas, and oil to generate electricity accounts for more than 30 percent of all polluting carbon dioxide.

3 Absorption

The remaining energy in sunlight is absorbed by the Earth's surface, converted into heat, and emitted into the atmosphere as long-wave, infrared radiation.

